Projection Learning and Conservative Projection Memory

Anonymous

Abstract

We introduce Projection Learning, a variational framework for runtime learning in which state updates are minimal consistency repairs. The update is a Bregman projection onto an evidence set, or its soft proximal analogue. We prove that projection is characterized by a Pythagorean principle: it is exactly the update that satisfies evidence while not increasing distance to any feasible ideal state. We also show that soft Projection Learning is implicit mirror descent, with ordinary gradient descent recovered as a linearized ℓ_2 special case. We instantiate the framework as Conservative Projection Memory, a neural associative state whose write direction is the innovation of a key relative to previously protected evidence. This yields an algebraic decomposition into redundant, novel, and contradictory evidence. We prove exact binding and capacity laws, conflict impossibility, no-free-repair lower bounds, optimal residual contraction, equivalence between sequential hard updates and batch projection, ridge competitive bounds, log-determinant information growth, and Bayesian interpretations. The resulting algorithms perform runtime adaptation by forward computation plus closed-form state repair, without inference-time backpropagation.

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1 Introduction

Backpropagation gives a clean mathematical primitive for train-time learning. A differentiable model produces an error, and the chain rule distributes that error through the computation graph. This is the right language when learning means changing static parameters to reduce a global objective.

Runtime learning has a different demand. A deployed system already has state: memories, estimates, cached structure, and commitments induced by previous observations. New evidence should change the state only where the evidence forces change. If the update is too weak, the system fails to adapt; if it is too broad, the system destroys information that was not contradicted. The basic problem is therefore not merely descent of a loss, but controlled repair of state under evidence.

Projection Learning formalizes this repair principle. Let S be a closed convex state space, let $\psi : S \to \mathbb{R}$ be differentiable and strictly convex, and define

$$D_\psi(u, v) = \psi(u) - \psi(v) - \langle \nabla\psi(v), u - v \rangle .$$

If evidence at time t gives a closed convex consistency set $\mathcal{C}_t \subseteq S$, the hard update is

$$s_{t+1} = \operatorname*{argmin}_{s \in \mathcal{C}_t} D_\psi(s, s_t).$$

If evidence is soft, represented by a convex loss $\ell_t : S \to \mathbb{R}$ and weight $\eta_t > 0$, the update is

$$s_{t+1} = \operatorname*{argmin}_{s \in S} \{D_\psi(s, s_t) + \eta_t \ell_t(s)\}.$$

The guiding question is simple: among the states that satisfy, or best approximate, the new evidence, which one changes the old state the least?

The canonical neural instance studied here is Conservative Projection Memory. A runtime memory is a matrix $W_t \in \mathbb{R}^{m \times r}$, and a hidden state produces a key $k_t \in \mathbb{R}^r$ and value $v_t \in \mathbb{R}^m$. The evidence asks for $Wk_t = v_t$. If old keys form K_{t-1}, the safe write direction is the innovation

$$z_t = (I - \Pi_{t-1})k_t, \qquad \Pi_{t-1} = K_{t-1}K_{t-1}^\dagger,$$

where $K_{t-1}^\dagger$ is the Moore-Penrose pseudoinverse. The soft conservative repair is

$$W_{t+1} = W_t + \frac{\eta_t}{1 + \eta_t \|z_t\|_2^2}(v_t - W_t k_t) z_t^\top .$$

Thus the memory does not write along the raw key. It writes only along the part of the key not already explained by protected evidence.

This innovation rule is the core intuition of the paper. Large innovation means the key contains a new direction. Small innovation with small residual means the evidence is redundant. Small innovation with large residual means the evidence conflicts with what the memory has committed to preserve. The theory below turns this intuition into exact feasibility laws, optimality statements, and online algorithms.

Organization. Section 2 fixes notation. Section 3 develops the main results as a sequence from projection geometry to conservative memory. Section 4 gives algorithms and complexity. Section 5 gives proof sketches. Section 6 concludes.

2 Preliminaries

2.1 Notation

All vector spaces are finite-dimensional over $\mathbb{R}$. Inner products are written $\langle x, y\rangle$. Vector lengths are written with explicit subscripts; in this paper they use the ℓ_2 norm $\|x\|_2$. For matrices, $\|A\|_F$ denotes the Frobenius norm, $\ker A$ denotes the nullspace, and $\operatorname{span}(A)$ denotes the span of the columns of A.

For a matrix K, the Moore-Penrose pseudoinverse is denoted by $K^\dagger$. The matrix $KK^\dagger$ is the orthogonal projector onto $\operatorname{span}(K)$. If $K = [k_1, \ldots, k_T] \in \mathbb{R}^{r\times T}$ and $V = [v_1, \ldots, v_T] \in \mathbb{R}^{m\times T}$, the equation $WK = V$ means that the memory $W \in \mathbb{R}^{m\times r}$ binds each key k_t to the corresponding value v_t.

2.2 Projection geometry

Let S be a closed convex subset of a finite-dimensional real vector space. Let $\psi : S \to \mathbb{R}$ be differentiable and strictly convex. The Bregman divergence generated by ψ is

$$D_\psi(u, v) = \psi(u) - \psi(v) - \langle \nabla\psi(v), u - v\rangle .$$

When $\psi(s) = \|s\|_2^2/2$, this is squared ℓ_2 distance. All projection statements below are finite-dimensional and use the standard variational characterization of convex minimizers.

2.3 Associative evidence

A memory state is a matrix $W \in \mathbb{R}^{m\times r}$. A collection of key-value evidence is written as

$$K = [k_1, \ldots, k_T] \in \mathbb{R}^{r\times T}, \qquad V = [v_1, \ldots, v_T] \in \mathbb{R}^{m\times T}.$$

Exact binding means $WK = V$. A new key $k \in \mathbb{R}^r$ is decomposed relative to the old key span by

$$\Pi = KK^\dagger, \qquad z = (I - \Pi)k, \qquad a = K^\dagger k. \tag{1}$$

The vector z is the innovation: the component of k not explained by previous keys. The coefficient vector a gives the least-squares representation of the old-span component of k.

2.4 Statement convention

The paper uses two levels of theorem statements. Main-text theorems are informal versions: they are mathematically precise enough to communicate the result and proof method, but omit some local regularity and typing details when those are fixed in the surrounding text. Each main-text theorem points to a formal appendix theorem. Each formal appendix theorem links back to its main-text version.

3 Main Results

This section states the main results in the order in which the argument is meant to be read. The first result identifies projection as the unique geometric form of stable repair. The next results explain why gradients still appear as a special case. The memory results then specialize the principle to associative state, where the notions of capacity, novelty, conflict, and optimal repair become explicit linear algebra.

3.1 Projection is forced by stability

Suppose new evidence determines a closed convex set $\mathcal{C}$ of acceptable states. A runtime learner should move from s_0 to a state in $\mathcal{C}$, but it should not disturb the state more than the evidence requires. The following theorem says that this intuitive requirement has an exact geometric form. Projection is equivalent to a Pythagorean inequality against every feasible ideal state.

Theorem 3.1 (Projection characterization, informal version of Theorem D.2)**.** *Let $\mathcal{C} \subseteq S$ be closed and convex. A point $p \in \mathcal{C}$ is the Bregman projection of s_0 onto $\mathcal{C}$ if and only if, for every $u \in \mathcal{C}$,*

$$D_\psi(u, p) + D_\psi(p, s_0) \leq D_\psi(u, s_0).$$

The left side has two terms: the remaining distance from the new state p to a feasible ideal u, and the price paid to move from s_0 to p. The theorem says that this sum is never larger than the original distance from s_0 to u. Thus projection is not just a way to satisfy evidence; it is the update that pays for its own movement by becoming closer to every state that could have explained the evidence.

3.2 Gradient learning is a linearized case

Projection Learning should not be read as an opposition to gradient descent. The relationship is more precise: soft projection is an implicit mirror step, and explicit gradient descent is obtained by replacing the new-state gradient by the old-state gradient. In this sense, gradient learning is the linearized local shadow of projection learning.

Theorem 3.2 (Soft projection as implicit mirror descent, informal version of Theorem D.5)**.** *Let ℓ_t be differentiable and suppose the soft projection optimum is interior:*

$$s_{t+1} = \operatorname*{argmin}_{s \in S} \{D_\psi(s, s_t) + \eta \ell_t(s)\}.$$

Then

$$\nabla\psi(s_{t+1}) = \nabla\psi(s_t) - \eta \nabla \ell_t(s_{t+1}).$$

For $\psi(s) = \|s\|_2^2/2$, this becomes implicit gradient descent. Replacing $\nabla\ell_t(s_{t+1})$ by $\nabla\ell_t(s_t)$ gives ordinary explicit gradient descent.

The distinction matters at runtime. A gradient step asks how the loss changes at the old state. A projection step asks which new state jointly stays near the old state and explains the evidence. For a quadratic consistency loss, this implicit update has an exact closed form.

Theorem 3.3 (Quadratic projection memory as exact implicit gradient descent, informal version of Theorem D.6)**.** *Let $k \in \mathbb{R}^r$, $v \in \mathbb{R}^m$, $W_t \in \mathbb{R}^{m \times r}$, and $\ell(W) = \|Wk - v\|_2^2/2$. The Frobenius soft projection update*

$$W_{t+1} = \operatorname*{argmin}_{W \in \mathbb{R}^{m \times r}} \left\{ \frac{1}{2} \|W - W_t\|_F^2 + \eta \ell(W) \right\}$$

is the implicit gradient step $W_{t+1} = W_t - \eta \nabla \ell(W_{t+1})$ and has the closed form

$$W_{t+1} = W_t + \frac{\eta}{1 + \eta \|k\|_2^2} (v - W_t k) k^\top.$$

Its residual contracts by $(1 + \eta \|k\|_2^2)^{-1}$ for all $\eta > 0$.

This theorem separates Projection Memory from naive fast-weight accumulation. The update is residual-correcting, normalized, and unconditionally contractive for the current evidence. The next question is what happens when there is more than one piece of evidence.

3.3 Binding has algebraic capacity

Associative memory begins with a set of desired bindings $Wk_i = v_i$. Stacking the keys and values gives the constraint $WK = V$. Not every such system is feasible: if the keys have a linear relation, the values must obey the same relation. The following theorem gives the exact feasibility condition and the closest feasible memory.

Theorem 3.4 (Exact associative binding, informal version of Theorem E.1)**.** *For key-value matrices $K \in \mathbb{R}^{r\times T}$ and $V \in \mathbb{R}^{m\times T}$, the constraint set $\{W : WK = V\}$ is nonempty if and only if $\ker K \subseteq \ker V$. When it is nonempty, the Frobenius projection of W_0 onto this set is*

$$W^\star = W_0 + (V - W_0K)K^\dagger.$$

In particular, if K has full column rank, then arbitrary values can be exactly bound.

The theorem gives a capacity law. An r-dimensional key space can bind arbitrary values for at most r linearly independent keys. After the key span is exhausted, additional values are not free choices; they are constrained by the linear relations already present in the keys. This is where runtime learning first encounters a hard boundary: some evidence is not new knowledge but a demand to violate previous structure.

3.4 Innovation is the safe write direction

Now imagine the memory already satisfies old evidence $WK = V$, and a new pair (k, v) arrives. Decompose the new key into the part already explained by old keys and the part outside their span:

$$k = Ka + z, \qquad a = K^\dagger k, \qquad z = (I - KK^\dagger)k.$$

The memory has freedom only on the innovation direction z. Everything in the old span is already fixed by V.

Theorem 3.5 (Innovation decomposition, informal version of Theorem E.2)**.** *Let $K \in \mathbb{R}^{r\times T}$, $V \in \mathbb{R}^{m\times T}$, $k \in \mathbb{R}^r$, and $v \in \mathbb{R}^m$. Assume old evidence is $WK = V$ and decompose the new key as in (1). Every old-feasible memory satisfies*

$$Wk = Va + Wz.$$

If $z = 0$, the value at k is already fixed by old evidence. If $z \neq 0$, the new evidence constrains only the innovation direction through $Wz = v - Va$.

This result is the hinge of the paper. It says that a safe runtime update should not write along the raw key k. The raw key contains old components that are already committed. The only direction that can carry new information without disturbing old constraints is z.

Theorem 3.6 (Conflict impossibility, informal version of Theorem E.3)**.** *Let $W_tK = V$ and let $z = (I - KK^\dagger)k$. If $z = 0$, then no update preserving $WK = V$ can change Wk. Hence evidence $Wk = v$ is compatible with old evidence if and only if $v = W_tk$.*

Thus conflict is not a training failure. It is an algebraic fact: the new constraint asks the memory to assign a different value to a key whose value is already implied by old evidence. A runtime learner should detect this case rather than blindly take a larger step.

3.5 Conservative repair is optimal

When $z \neq 0$, the new evidence can be repaired while preserving old constraints. The soft version trades exact satisfaction for controlled movement. It is the proximal repair of the current residual inside the old-feasible affine subspace.

Theorem 3.7 (Soft Conservative Projection Memory, informal version of Theorem E.8)**.** *Let $K \in \mathbb{R}^{r\times T}$, $V \in \mathbb{R}^{m\times T}$, $W_t \in \mathbb{R}^{m\times r}$, $k \in \mathbb{R}^r$, and $v \in \mathbb{R}^m$. Assume $W_t K = V$, define $z = (I - KK^\dagger)k$, and suppose $z \neq 0$. The constrained soft repair problem*

$$W_{t+1} = \operatorname*{argmin}_{W:WK=V} \left\{ \frac{1}{2} \|W - W_t\|_F^2 + \frac{\eta}{2} \|Wk - v\|_2^2 \right\}$$

has the closed form

$$W_{t+1} = W_t + \frac{\eta}{1 + \eta \|z\|_2^2} (v - W_t k) z^\top.$$

It preserves old constraints and contracts the new residual by the factor $(1 + \eta \|z\|_2^2)^{-1}$.

The denominator is not an implementation trick. It is the price of implicit repair in the innovation direction. Large innovation permits a strong contraction with modest movement. Small innovation means the key is almost already explained by previous evidence, so a large residual is difficult to repair safely.

Theorem 3.8 (No-free-repair lower bound, informal version of Theorem E.5)**.** *Let $K \in \mathbb{R}^{r\times T}$, $V \in \mathbb{R}^{m\times T}$, $W_t \in \mathbb{R}^{m\times r}$, $k \in \mathbb{R}^r$, and $v \in \mathbb{R}^m$. Assume $W_t K = V$, $\Pi = KK^\dagger$, $z = (I - \Pi)k$, and $e = v - W_t k$. Let an update $W_t + \Delta$ preserve old constraints, so $\Delta K = 0$. If the post-update residual obeys*

$$\|v - (W_t + \Delta)k\|_2 \leq \gamma \|e\|_2$$

for $0 \leq \gamma < 1$ and $z \neq 0$, then

$$\|\Delta\|_F \geq \frac{(1-\gamma)\|e\|_2}{\|z\|_2}.$$

If $z = 0$ and $e \neq 0$, no such contraction is possible while preserving old constraints.

This lower bound explains the novelty-conflict diagnostic. Small $\|z\|_2$ and large $\|e\|_2$ are not merely bad for Conservative Projection Memory; they are intrinsically expensive for any update that preserves old evidence.

Theorem 3.9 (Optimal contraction, informal version of Theorem E.6)**.** *Under the notation of Theorem 3.8, fix $0 \leq \gamma \leq 1$ and suppose $z \neq 0$. Among all updates preserving $\Delta K = 0$ and achieving exact residual $v - (W_t + \Delta)k = \gamma e$, the minimum-Frobenius-norm update is*

$$\Delta^\star = (1-\gamma) \frac{e z^\top}{\|z\|_2^2}.$$

It attains the lower bound in Theorem 3.8. Soft CPM is the special case $\gamma = (1 + \eta \|z\|_2^2)^{-1}$.

Together, the lower bound and the optimality theorem say that Conservative Projection Memory is not merely conservative; it is as conservative as possible for its chosen amount of residual repair.

3.6 The online hard update is the batch projection

The hard conservative update exactly satisfies each compatible new constraint. A natural worry is that doing this sequentially might be greedy and order-dependent. The next theorem removes that worry in the compatible case: the sequential hard procedure returns the same memory as the batch projection onto all evidence.

Theorem 3.10 (Sequential hard CPM equals batch projection, informal version of Theorem E.7)**.** *For compatible evidence $(k_t, v_t)_{t=1}^T$, sequential hard CPM with redundant steps skipped returns*

$$W_{T+1} = \operatorname*{argmin}_{W:WK_T=V_T} \frac{1}{2} \|W - W_0\|_F^2 = W_0 + (V_T - W_0 K_T) K_T^\dagger.$$

Thus the online hard update exactly tracks the batch Frobenius projection solution.

This result completes the exact story. Projection gives the closest memory satisfying all compatible evidence; innovation tells the online learner where the next constraint has new degrees of freedom; hard CPM follows those degrees of freedom and arrives at the same point as the batch solution.

3.7 Ridge memory turns exact repair into statistical repair

Exact preservation is appropriate when evidence is trusted and the protected constraints should not move. Long noisy streams require a softer object: evidence should accumulate with weights, old information should regularize the state, and uncertain directions should be updated cautiously. Ridge Projection Memory is that statistical version.

Theorem 3.11 (Ridge and Bayesian projection memory, informal version of Theorem F.5)**.** *Let $W_0 \in \mathbb{R}^{m\times r}$, $\lambda > 0$, $\rho_\tau \geq 0$, $k_\tau \in \mathbb{R}^r$, and $v_\tau \in \mathbb{R}^m$. The ridge objective*

$$J_t(W) = \frac{\lambda}{2} \|W - W_0\|_F^2 + \frac{1}{2} \sum_{\tau=1}^{t} \rho_\tau \|W k_\tau - v_\tau\|_2^2$$

has minimizer $W_{t+1} = B_t A_t^{-1}$, where

$$A_t = \lambda I + \sum_{\tau=1}^{t} \rho_\tau k_\tau k_\tau^\top, \qquad B_t = \lambda W_0 + \sum_{\tau=1}^{t} \rho_\tau v_\tau k_\tau^\top.$$

The same matrix is the posterior mean and MAP estimate under a Gaussian linear observation model.

The inverse covariance in ridge memory plays the role that the protected subspace plays in exact CPM. It remembers which key directions have already been supported by evidence and scales future updates accordingly.

Theorem 3.12 (Ridge bound, informal version of Theorem F.3)**.** *Under the notation of Theorem 3.11, let the pre-update ridge state and covariance be*

$$W_t = \operatorname*{argmin}_{W\in\mathbb{R}^{m\times r}} J_{t-1}(W), \qquad P_t = \left(\lambda I + \sum_{\tau<t} \rho_\tau k_\tau k_\tau^\top\right)^{-1}.$$

Define $a_t = k_t^\top P_t k_t$. For every comparator U,

$$\sum_{t=1}^{T} \frac{\rho_t}{1 + \rho_t a_t} \|v_t - W_t k_t\|_2^2 \leq \lambda \|U - W_0\|_F^2 + \sum_{t=1}^{T} \rho_t \|v_t - U k_t\|_2^2.$$

The normalized loss weights errors by the leverage a_t. A direction that is uncertain has high leverage and receives a cautious normalized penalty; a direction that has become well supported is judged more directly. The corresponding information growth is exactly log-determinantal.

Theorem 3.13 (Log-determinant information identity, informal version of Theorem F.4)**.** *Let* $\lambda > 0$, $\rho_t \geq 0$, $k_t \in \mathbb{R}^r$, $A_t = \lambda I + \sum_{\tau=1}^{t} \rho_\tau k_\tau k_\tau^\top$, *and* $a_t = k_t^\top A_{t-1}^{-1} k_t$. *Then*

$$\sum_{t=1}^{T} \log(1 + \rho_t a_t) = \log \frac{\det A_T}{\det A_0}.$$

This identity gives the statistical version of the capacity story. Exact CPM counts independent innovation directions; ridge memory measures accumulated evidence volume through the log determinant of the key covariance.

4 Algorithms and Complexity

The theory becomes an algorithm once the learner keeps two kinds of objects separate. The slow parameters are the backbone and the maps K, V, B that decide which features become keys, values, and memory readouts. These parameters are trained by ordinary backpropagation. The fast state is the runtime memory. It is updated by the same projection rule during training and inference.

At each step the algorithm asks two questions. The innovation ℓ_2 norm asks whether the current key contains a direction not protected by old evidence. The residual ℓ_2 norm asks whether the current memory already predicts the value associated with that key. Together they distinguish novelty, redundancy, and conflict.

4.1 Streaming Conservative Projection Memory

The exact conservative algorithm maintains a memory $W_t \in \mathbb{R}^{m \times r}$ and an orthonormal basis $Q_t \in \mathbb{R}^{r \times q_t}$ for protected key directions. The projector $Q_t Q_t^\top$ is the computational form of the old key span.

- **Part 1. Read.** Given $h_t \in \mathbb{R}^d$, compute

  $$k_t = K h_t, \qquad v_t = V h_t, \qquad r_t = W_t k_t, \qquad \tilde{h}_t = h_t + B r_t.$$

  The model first reads from the old state. This read-before-write convention prevents the current value from being immediately copied into its own prediction.

- **Part 2. Diagnose.** Compute

  $$z_t = k_t - Q_t Q_t^\top k_t, \qquad e_t = v_t - W_t k_t.$$

  Here $\|z_t\|_2$ measures novelty of the key direction, while $\|e_t\|_2$ measures surprise of the value under the current memory.

- **Part 3. Repair.** If $\|z_t\|_2^2 > \varepsilon$, update

  $$W_{t+1} = W_t + \frac{\eta_t}{1 + \eta_t \|z_t\|_2^2} e_t z_t^\top$$

  and optionally append $z_t / \|z_t\|_2$ to Q_t. If $\|z_t\|_2^2 \leq \varepsilon$ and $\|e_t\|_2$ is small, the evidence is redundant. If $\|z_t\|_2^2 \leq \varepsilon$ and $\|e_t\|_2$ is large, the evidence is conflicting relative to protected memory and cannot be safely repaired without relaxing old commitments.

The exact algorithm is useful when the system should protect a finite set of remembered directions. Its per-step overhead is

$$O(d(r+m) + dm + mr + rq_t).$$

If $m = r$ and $q_t \leq r$, this becomes

$$O(dr + mr + r^2).$$

The runtime state cost is

$$O(mr + rq_t).$$

Since at most r independent key directions can be protected exactly, $q_t \leq r$ in the exact finite-rank regime.

4.2 Ridge Projection Memory

When evidence is noisy or the stream is long, exact preservation can be too rigid. Ridge Projection Memory replaces the protected subspace by an inverse covariance P_t. Directions with strong accumulated key evidence get smaller future gains; unsupported directions remain plastic.

The ridge algorithm maintains $W_t \in \mathbb{R}^{m \times r}$ and $P_t \in \mathbb{R}^{r \times r}$, initialized by

$$W_1 = W_0, \qquad P_1 = \lambda^{-1} I.$$

At step t, after computing $k_t, v_t, r_t, \tilde{h}_t$, choose $\rho_t \geq 0$ and update

$$\begin{aligned} g_t &= \frac{\rho_t P_t k_t}{1 + \rho_t k_t^\top P_t k_t}, \\ W_{t+1} &= W_t + (v_t - W_t k_t) g_t^\top, \\ P_{t+1} &= P_t - \frac{\rho_t P_t k_t k_t^\top P_t}{1 + \rho_t k_t^\top P_t k_t}. \end{aligned}$$

This is the recursive form of the ridge projection solution and also the Bayesian posterior update for the linear-Gaussian memory model.

The per-step overhead is

$$O(d(r+m) + dm + mr + r^2),$$

and the runtime state cost is

$$O(mr + r^2).$$

For small r and m, the update is a forward pass plus low-rank state repair rather than an inference-time backward pass.

4.3 Training and inference

The same state-update rule is used during training and inference. Training learns the slow maps that make projection useful; inference uses the learned maps and continues to update only the fast state.

- **Part 1. Fully unrolled training.** Projection updates are unrolled and differentiated through. This gives K, V, B direct long-horizon credit for how their keys and values shape future memory, at the cost of storing runtime states across the sequence.
- **Part 2. Truncated state-gradient training.** Gradients are propagated through only the most recent H state updates, giving state memory $O(HM_s)$ for runtime-state size M_s. This is the natural analogue of truncated backpropagation through time.
- **Part 3. Detached-state training.** Projection updates are executed with stop-gradient. This has inference-like memory cost. Since long-horizon credit to V is weakened, one may add local consistency losses or occasionally unroll state gradients.

At inference, there is no backward pass. The cost per token is

$$O(C_{\text{bb}} + C_{\text{pm}}),$$

where C_{bb} is the backbone forward cost and C_{pm} is the projection-memory overhead above. In contrast, TTT-style adaptation typically requires an inner backward pass through the test-time module. Projection memory is forward computation plus closed-form rank-one state repair.

5 Technical Overview

This section sketches the proof ideas. Full formal statements and proofs appear in Appendices D, E, and F.

Theorem 5.1 (Projection characterization, proof sketch for Theorem D.2)**.** *The informal statement is Theorem 3.1.*

Proof sketch. The optimality condition for $p = \operatorname{argmin}_{s \in \mathcal{C}} D_\psi(s, s_0)$ is

$$\langle \nabla\psi(p) - \nabla\psi(s_0), u - p \rangle \geq 0 \qquad \forall u \in \mathcal{C}.$$

The Bregman three-point identity gives

$$D_\psi(u, s_0) - D_\psi(u, p) - D_\psi(p, s_0) = \langle \nabla\psi(s_0) - \nabla\psi(p), p - u \rangle .$$

The two displays are equivalent. This proves both directions. □

Theorem 5.2 (Implicit mirror descent, proof sketch for Theorem D.5)**.** *The informal statement is Theorem 3.2.*

Proof sketch. Differentiate the soft projection objective at the interior optimum:

$$0 = \nabla_s D_\psi(s_{t+1}, s_t) + \eta \nabla \ell_t(s_{t+1}).$$

Since

$$\nabla_s D_\psi(s, s_t) = \nabla\psi(s) - \nabla\psi(s_t),$$

we obtain

$$\nabla\psi(s_{t+1}) = \nabla\psi(s_t) - \eta \nabla \ell_t(s_{t+1}).$$

□

Theorem 5.3 (Quadratic memory and implicit gradient descent, proof sketch for Theorem D.6)**.** *The informal statement is Theorem 3.3.*

Proof sketch. For $\ell(W) = \|Wk - v\|_2^2/2$,

$$\nabla \ell(W) = (Wk - v)k^\top.$$

The implicit step is

$$W_{t+1} = W_t + \eta(v - W_{t+1}k)k^\top.$$

With $e = v - W_t k$ and $W_{t+1} = W_t + \alpha e k^\top$,

$$v - W_{t+1}k = (1 - \alpha \|k\|_2^2)e.$$

Comparing coefficients gives

$$\alpha = \eta(1 - \alpha \|k\|_2^2) = \frac{\eta}{1 + \eta \|k\|_2^2}.$$

□

Theorem 5.4 (Exact binding, proof sketch for Theorem E.1)**.** *The informal statement is Theorem 3.4.*

Proof sketch. If $WK = V$ and $Ka = 0$, then

$$Va = WKa = 0.$$

Thus $\ker K \subseteq \ker V$. Conversely this nullspace condition implies

$$V = VK^\dagger K,$$

so $W = VK^\dagger$ is feasible. For projection, write $W = W_0 + \Delta$ and $R = V - W_0 K$. The problem becomes

$$\min_\Delta \frac{1}{2} \|\Delta\|_F^2 \quad \text{subject to} \quad \Delta K = R.$$

The minimum-Frobenius-norm feasible solution is $\Delta = RK^\dagger$. □

Theorem 5.5 (Innovation and conflict, proof sketch for Theorems E.2 and E.3)**.** *The informal statements are Theorems 3.5 and 3.6.*

Proof sketch. Using (1),

$$k = KK^\dagger k + (I - KK^\dagger)k = Ka + z.$$

If $WK = V$, then

$$Wk = WKa + Wz = Va + Wz.$$

If $z = 0$, then $Wk = Va$ for every old-feasible W. If an update Δ preserves old evidence, then $\Delta K = 0$ and hence $\Delta k = 0$ whenever $k \in \operatorname{span}(K)$. Thus contradiction cannot be repaired without breaking old evidence. □

Theorem 5.6 (Soft CPM, proof sketch for Theorem E.8)**.** *The informal statement is Theorem 3.7.*

Proof sketch. Set $W = W_t + \Delta$ and $e = v - W_t k$. Since $W_t K = V$, preserving old evidence is $\Delta K = 0$. Because $\Delta K = 0$,

$$\Delta k = \Delta(I - \Pi)k = \Delta z.$$

The objective becomes

$$\min_{\Delta:\Delta K=0} \frac{1}{2} \|\Delta\|_F^2 + \frac{\eta}{2} \|\Delta z - e\|_2^2 .$$

The stationarity equation inside this subspace is

$$\Delta + \eta(\Delta z - e)z^\top = 0.$$

Solving with $\Delta = \alpha e z^\top$ yields

$$\alpha = \frac{\eta}{1 + \eta \|z\|_2^2}.$$

The residual satisfies

$$v - W_{t+1}k = e - \Delta z = \frac{1}{1 + \eta \|z\|_2^2} e.$$

□

Theorem 5.7 (No-free repair and optimal contraction, proof sketch for Theorems E.5 and E.6)**.** *The informal statements are Theorems 3.8 and 3.9.*

Proof sketch. Every old-preserving update satisfies

$$\Delta k = \Delta(I - \Pi)k = \Delta z.$$

The post-update residual is

$$e^+ = e - \Delta z.$$

If $\|e^+\|_2 \le \gamma \|e\|_2$, then

$$\|\Delta z\|_2 = \left\|e - e^+\right\|_2 \ge (1 - \gamma) \|e\|_2 .$$

Since $\|\Delta z\|_2 \le \|\Delta\|_F \|z\|_2$, the lower bound follows. If one demands $e^+ = \gamma e$, then $\Delta z = (1 - \gamma)e$, and the minimum-Frobenius-norm solution is

$$\Delta^\star = (1 - \gamma) \frac{e z^\top}{\|z\|_2^2}.$$

□

Theorem 5.8 (Sequential hard CPM, proof sketch for Theorem E.7)**.** *The informal statement is Theorem 3.10.*

Proof sketch. Let $\mathcal{C}_t = \{W : WK_t = V_t\}$ and $\mathcal{T}_t = \{H : HK_t = 0\}$. By induction, $W_t \in \mathcal{C}_{t-1}$ and

$$\langle H, W_t - W_0 \rangle_F = 0 \qquad \forall H \in \mathcal{T}_{t-1}.$$

The hard CPM update lies in $\mathcal{C}_t$. If $H \in \mathcal{T}_t$, then $Hz_t = 0$, so

$$\left\langle H, e_t z_t^\top \right\rangle_F = e_t^\top H z_t = 0.$$

Thus $W_{t+1} - W_0$ is orthogonal to $\mathcal{T}_t$, which is the Frobenius projection condition for $\mathcal{C}_t$. □

Theorem 5.9 (Ridge and Bayesian memory, proof sketch for Theorem F.5)**.** *The informal statement is Theorem 3.11.*

Proof sketch. Differentiating J_t gives

$$0 = \lambda(W - W_0) + \sum_{\tau=1}^{t} \rho_\tau (Wk_\tau - v_\tau) k_\tau^\top.$$

Rearranging,

$$W \left(\lambda I + \sum_{\tau=1}^{t} \rho_\tau k_\tau k_\tau^\top \right) = \lambda W_0 + \sum_{\tau=1}^{t} \rho_\tau v_\tau k_\tau^\top.$$

Thus $W = B_t A_t^{-1}$. Under the Gaussian prior and observation model, the negative log posterior equals $J_t(W)$ up to constants, so the posterior mean and MAP coincide with this minimizer. □

Theorem 5.10 (Ridge competitive bound and log determinant, proof sketch for Theorems F.3 and F.4)**.** *The informal statements are Theorems 3.12 and 3.13.*

Proof sketch. Around the minimizer W_t of J_{t-1}, the increment minimization is

$$\min_{\Delta} \frac{1}{2} \operatorname{tr}(\Delta A_{t-1} \Delta^\top) + \frac{\rho_t}{2} \|\Delta k_t - e_t\|_2^2.$$

Its minimum value is

$$\frac{\rho_t}{2(1 + \rho_t a_t)} \|e_t\|_2^2.$$

Summing and comparing $J_T(W_{T+1})$ with $J_T(U)$ gives the competitive bound. The log-determinant identity follows from

$$\det(A_{t-1} + \rho_t k_t k_t^\top) = \det(A_{t-1})(1 + \rho_t a_t).$$

□

6 Conclusion

The paper begins from a simple mismatch. Backpropagation is a powerful primitive for training static differentiable functions, but a runtime learner faces a different demand: it must absorb evidence while preserving what the evidence does not force it to change. Projection Learning formalizes this demand as minimal consistency repair.

The mathematical arc is that projection is characterized by stability, soft projection contains implicit mirror descent, and gradient descent appears as a linearized ℓ_2 case. Conservative Projection Memory then realizes the principle in a neural associative state. Its central rule is to write along the innovation of a key rather than the key itself. This single change exposes a complete algebra of runtime evidence: exact capacity, redundancy, novelty, contradiction, lower bounds on safe repair, optimal residual contraction, and equivalence between sequential hard repair and batch projection.

The broader conclusion is that runtime learning should make the geometry of evidence visible. A useful adaptive system should be able to say what is already implied, what is genuinely new, what is statistically uncertain, and what cannot be changed without breaking prior commitments. Projection memory supplies one concrete mechanism for doing so with forward computation and closed-form state repair, without inference-time backpropagation.

References

[AW01] Katy S. Azoury and Manfred K. Warmuth. Relative loss bounds for on-line density estimation with the exponential family of distributions. *Machine Learning*, 43:211–246, 2001.

[BHM+16] Jimmy Ba, Geoffrey E. Hinton, Volodymyr Mnih, Joel Z. Leibo, and Catalin Ionescu. Using fast weights to attend to the recent past. In *Advances in Neural Information Processing Systems*, volume 29, 2016.

[Bre67] Lev M. Bregman. The relaxation method of finding the common point of convex sets and its application to the solution of problems in convex programming. *USSR Computational Mathematics and Mathematical Physics*, 7(3):200–217, 1967.

[CDK+06] Koby Crammer, Ofer Dekel, Joseph Keshet, Shai Shalev-Shwartz, and Yoram Singer. Online passive-aggressive algorithms. *Journal of Machine Learning Research*, 7:551–585, 2006.

[Haz16] Elad Hazan. *Introduction to Online Convex Optimization*. Foundations and Trends in Optimization, 2016.

[JCO+17] Max Jaderberg, Wojciech Marian Czarnecki, Simon Osindero, Oriol Vinyals, Alex Graves, David Silver, and Koray Kavukcuoglu. Decoupled neural interfaces using synthetic gradients. In *Proceedings of the 34th International Conference on Machine Learning*, volume 70 of *Proceedings of Machine Learning Research*, pages 1627–1635, 2017.

[KVPF20] Angelos Katharopoulos, Apoorv Vyas, Nikolaos Pappas, and François Fleuret. Transformers are RNNs: Fast autoregressive transformers with linear attention. In *Proceedings of the 37th International Conference on Machine Learning*, volume 119 of *Proceedings of Machine Learning Research*, pages 5156–5165, 2020.

[RHW86] David E. Rumelhart, Geoffrey E. Hinton, and Ronald J. Williams. Learning representations by back-propagating errors. *Nature*, 323(6088):533–536, 1986.

[SIS21] Imanol Schlag, Kazuki Irie, and Jürgen Schmidhuber. Linear transformers are secretly fast weight programmers. In *Proceedings of the 38th International Conference on Machine Learning*, volume 139 of *Proceedings of Machine Learning Research*, pages 9355–9366, 2021.

[SLD+25] Yu Sun, Xinhao Li, Karan Dalal, Jiarui Xu, Arjun Vikram, Genghan Zhang, Yann Dubois, Xinlei Chen, Xiaolong Wang, Sanmi Koyejo, Tatsunori Hashimoto, and Carlos Guestrin. Learning to (learn at test time): RNNs with expressive hidden states. In *Proceedings of the 42nd International Conference on Machine Learning*, volume 267 of *Proceedings of Machine Learning Research*, pages 57503–57522, 2025.

[SS12] Shai Shalev-Shwartz. Online learning and online convex optimization. *Foundations and Trends in Machine Learning*, 4(2):107–194, 2012.

[SWL+20] Yu Sun, Xiaolong Wang, Zhuang Liu, John Miller, Alexei A. Efros, and Moritz Hardt. Test-time training with self-supervision for generalization under distribution shifts. In *Proceedings of the 37th International Conference on Machine Learning*, volume 119 of *Proceedings of Machine Learning Research*, pages 9229–9248, 2020.

[Vov97] Volodya Vovk. Competitive on-line linear regression. In *Advances in Neural Information Processing Systems*, volume 10, 1997.

[Zin03] Martin Zinkevich. Online convex programming and generalized infinitesimal gradient ascent. In *Proceedings of the 20th International Conference on Machine Learning*, pages 928–936, 2003.

Appendix

A Appendix Roadmap

The appendix follows the same chain as the main text, but with every informal statement replaced by a formal theorem and proof. It first states what the theory does not claim, then places the work relative to existing methods, and then develops the proofs from general projection geometry to exact memory and finally to ridge-statistical memory.

- **Part 1.** Appendix B states the limitations of the framework and clarifies which questions remain empirical or architectural rather than settled by the present theory.
- **Part 2.** Appendix C situates Projection Learning among backpropagation, mirror descent, passive-aggressive learning, fast weights, recursive least squares, TTT-style runtime adaptation, and information projection.
- **Part 3.** Appendix D proves the general geometry: the Bregman identity, the projection characterization theorem, soft projection inequalities, KL projection, and the relation to implicit and explicit gradient learning.
- **Part 4.** Appendix E proves the exact memory story: feasibility, capacity, innovation decomposition, conflict impossibility, conservative hard and soft updates, no-free-repair lower bounds, optimal contraction, and equivalence between sequential hard CPM and batch projection.
- **Part 5.** Appendix F proves the statistical memory story: recursive ridge updates, Bayesian interpretation, normalized competitive prediction, and log-determinant information growth.

B Limitations

Projection Learning is a formal update principle, not a guarantee that every source of evidence is useful. If the value map Vh_t produces a poor target, projection will repair the wrong constraint. The theory also separates feasibility from generalization: exact binding says when constraints can be satisfied, but it does not imply that the resulting memory improves downstream prediction.

Conservative Projection Memory assumes that preserving old constraints is desirable. This is appropriate for stable binding, but nonstationary environments may require forgetting, reset, branching, or discounted ridge objectives. Exact innovation maintenance also has a computational cost. In large systems, one may use rank caps, approximate orthogonalization, diagonal or low-rank covariance estimates, or learned gates; these approximations are not covered by the exact preservation theorems.

The paper proves deterministic algebraic, convex-analytic, and online regression statements. It does not prove end-to-end performance improvements for language modeling, classification under shift, or long-context retrieval. Those claims require experiments. The proposed diagnostics are residual, innovation norm, conflict rate, leverage, and contraction ratio; the theory predicts their behavior under the stated model, not their empirical distribution in trained networks.

C Related Work

Backpropagation and runtime updates. Backpropagation supplies differentiable credit assignment for static computation graphs [RHW86]. Synthetic gradients decouple modules by predicting downstream gradient information [JCO+17]. Projection Learning decouples runtime updates differently: the update is not a predicted derivative, but the solution of a local state-repair problem.

Bregman projection and online optimization. Bregman projections originate in convex feasibility and optimization [Bre67]. Online convex optimization, mirror descent, and proximal methods use related geometries to bound regret and movement [Zin03, SS12, Haz16]. The present paper uses this geometry as a runtime-learning semantics: evidence defines consistency, and projection defines the least disruptive repair. The implicit mirror-descent view clarifies that gradient descent is not rejected, but recovered as a linearized ℓ_2 special case.

Passive-aggressive learning and least squares. Passive-aggressive algorithms solve minimal-change constrained updates for supervised online learning [CDK+06]. Recursive least squares and online ridge regression yield classical closed-form updates and loss bounds [Vov97, AW01]. Conservative Projection Memory adapts these ideas to internal neural state and adds the innovation/conflict decomposition induced by old evidence.

Test-time training and expressive hidden states. Test-time training updates model parameters on self-supervised objectives at inference time [SWL+20]. TTT layers make hidden states themselves trainable models updated on test sequences [SLD+25]. Conservative Projection Memory targets the same need for runtime adaptation, but replaces test-time backpropagation with closed-form projection updates.

Fast weights and linear attention. Fast weights store temporary information in rapidly changing associative matrices [BHM+16]. Linear attention can be interpreted through recurrent fast-weight updates [KVPF20, SIS21]. Conservative Projection Memory differs by writing residuals rather than raw values and by writing in the key innovation subspace rather than along the raw key.

D Projection Learning Proofs

Lemma D.1 (Bregman three-point identity)**.** *For all $a, b, u \in S$,*

$$D_\psi(u, a) - D_\psi(u, b) - D_\psi(b, a) = \langle \nabla\psi(a) - \nabla\psi(b), b - u \rangle .$$

Proof. Expand each term:

$$\begin{aligned} D_\psi(u, a) &= \psi(u) - \psi(a) - \langle \nabla\psi(a), u - a \rangle , \\ D_\psi(u, b) &= \psi(u) - \psi(b) - \langle \nabla\psi(b), u - b \rangle , \\ D_\psi(b, a) &= \psi(b) - \psi(a) - \langle \nabla\psi(a), b - a \rangle . \end{aligned}$$

Then

$$D_\psi(u, a) - D_\psi(u, b) - D_\psi(b, a) = - \langle \nabla\psi(a), u - a \rangle + \langle \nabla\psi(b), u - b \rangle + \langle \nabla\psi(a), b - a \rangle$$

$$
\begin{aligned}
&= \langle \nabla\psi(a), b-u\rangle - \langle \nabla\psi(b), b-u\rangle \\
&= \langle \nabla\psi(a) - \nabla\psi(b), b-u\rangle .
\end{aligned}
$$

□

Theorem D.2 (Projection characterization, formal version of Theorem 3.1)**.** *Let S be closed and convex, let ψ be differentiable and strictly convex, let $\mathcal{C} \subseteq S$ be closed and convex, and let $s_0 \in S$. For $p \in \mathcal{C}$, the following are equivalent:*

- **Part 1.** $p = \operatorname{argmin}_{s\in\mathcal{C}} D_\psi(s, s_0)$.
- **Part 2.** *For all $u \in \mathcal{C}$,*

$$
D_\psi(u,p) + D_\psi(p,s_0) \le D_\psi(u,s_0).
$$

Proof. Assume Part 1. The first-order optimality condition is

$$
\langle \nabla\psi(p) - \nabla\psi(s_0), u-p\rangle \ge 0 \qquad \forall u \in \mathcal{C}.
$$

Equivalently,

$$
\langle \nabla\psi(s_0) - \nabla\psi(p), p-u\rangle \ge 0 \qquad \forall u \in \mathcal{C}.
$$

By Lemma D.1,

$$
D_\psi(u,s_0) - D_\psi(u,p) - D_\psi(p,s_0) = \langle \nabla\psi(s_0) - \nabla\psi(p), p-u\rangle .
$$

Thus Part 2 follows.

Assume Part 2. The same identity implies

$$
\langle \nabla\psi(s_0) - \nabla\psi(p), p-u\rangle \ge 0 \qquad \forall u \in \mathcal{C}.
$$

Hence

$$
\langle \nabla\psi(p) - \nabla\psi(s_0), u-p\rangle \ge 0 \qquad \forall u \in \mathcal{C}.
$$

This is the variational inequality for minimizing $D_\psi(s, s_0)$ over $\mathcal{C}$. Strict convexity gives uniqueness, so Part 1 follows. □

Corollary D.3 (Fejer monotonicity and bounded drift)**.** *Let $s_{t+1} = \operatorname{argmin}_{s\in\mathcal{C}_t} D_\psi(s, s_t)$. If $u^\star \in \cap_{t=1}^T \mathcal{C}_t$, then*

$$
D_\psi(u^\star, s_{t+1}) \le D_\psi(u^\star, s_t)
$$

and

$$
\sum_{t=1}^{T} D_\psi(s_{t+1}, s_t) \le D_\psi(u^\star, s_1).
$$

Proof. By Theorem D.2,

$$D_\psi(u^\star, s_{t+1}) + D_\psi(s_{t+1}, s_t) \le D_\psi(u^\star, s_t).$$

Dropping the second term gives the first claim. Summing gives

$$\begin{aligned}\sum_{t=1}^{T} D_\psi(s_{t+1}, s_t) &\le \sum_{t=1}^{T} \left(D_\psi(u^\star, s_t) - D_\psi(u^\star, s_{t+1})\right) \\ &= D_\psi(u^\star, s_1) - D_\psi(u^\star, s_{T+1}) \\ &\le D_\psi(u^\star, s_1).\end{aligned}$$

□

Theorem D.4 (Soft projection inequality)**.** *Let $\ell_t : S \to \mathbb{R}$ be convex and let*

$$s_{t+1} = \operatorname*{argmin}_{s \in S} \{D_\psi(s, s_t) + \eta_t \ell_t(s)\}.$$

Then for all $u \in S$,

$$\eta_t(\ell_t(s_{t+1}) - \ell_t(u)) \le D_\psi(u, s_t) - D_\psi(u, s_{t+1}) - D_\psi(s_{t+1}, s_t).$$

Proof. For some $g_{t+1} \in \partial \ell_t(s_{t+1})$, optimality gives

$$\langle \nabla\psi(s_{t+1}) - \nabla\psi(s_t) + \eta_t g_{t+1}, u - s_{t+1} \rangle \ge 0.$$

Thus

$$\eta_t \langle g_{t+1}, s_{t+1} - u \rangle \le \langle \nabla\psi(s_t) - \nabla\psi(s_{t+1}), s_{t+1} - u \rangle .$$

Convexity gives

$$\ell_t(s_{t+1}) - \ell_t(u) \le \langle g_{t+1}, s_{t+1} - u \rangle .$$

Lemma D.1 gives

$$\langle \nabla\psi(s_t) - \nabla\psi(s_{t+1}), s_{t+1} - u \rangle = D_\psi(u, s_t) - D_\psi(u, s_{t+1}) - D_\psi(s_{t+1}, s_t).$$

Combining proves the claim. □

Theorem D.5 (Soft projection as implicit mirror descent, formal version of Theorem 3.2)**.** *Let ℓ_t be differentiable and let*

$$s_{t+1} = \operatorname*{argmin}_{s \in S} \{D_\psi(s, s_t) + \eta \ell_t(s)\}.$$

If s_{t+1} is an interior optimum, then

$$\nabla\psi(s_{t+1}) = \nabla\psi(s_t) - \eta \nabla \ell_t(s_{t+1}).$$

If $\psi(s) = \|s\|_2^2/2$, then

$$s_{t+1} = s_t - \eta \nabla \ell_t(s_{t+1}).$$

Proof. At an interior optimum,

$$0 = \nabla_s D_\psi(s_{t+1}, s_t) + \eta \nabla \ell_t(s_{t+1}).$$

Since

$$\nabla_s D_\psi(s, s_t) = \nabla \psi(s) - \nabla \psi(s_t),$$

we obtain

$$\nabla \psi(s_{t+1}) - \nabla \psi(s_t) + \eta \nabla \ell_t(s_{t+1}) = 0.$$

Rearranging proves the mirror form. If $\psi(s) = \|s\|_2^2/2$, then $\nabla \psi(s) = s$. □

Theorem D.6 (Quadratic projection memory as implicit gradient descent, formal version of Theorem 3.3)**.** *Let $k \in \mathbb{R}^r$, $v \in \mathbb{R}^m$, $W_t \in \mathbb{R}^{m \times r}$, $\eta > 0$, and $\ell(W) = \|Wk - v\|_2^2/2$. The update*

$$W_{t+1} = \operatorname*{argmin}_{W \in \mathbb{R}^{m \times r}} \left\{ \frac{1}{2} \|W - W_t\|_F^2 + \eta \ell(W) \right\}$$

is equivalent to

$$W_{t+1} = W_t - \eta \nabla \ell(W_{t+1})$$

and has closed form

$$W_{t+1} = W_t + \frac{\eta}{1 + \eta \|k\|_2^2} (v - W_t k) k^\top.$$

Moreover, if $e = v - W_t k$, then

$$v - W_{t+1} k = \frac{1}{1 + \eta \|k\|_2^2} e.$$

Proof. The gradient is

$$\nabla \ell(W) = (Wk - v) k^\top.$$

The optimality condition for the proximal objective is

$$0 = W_{t+1} - W_t + \eta (W_{t+1} k - v) k^\top,$$

which is the implicit gradient step. Let $e = v - W_t k$ and set $W_{t+1} = W_t + \alpha e k^\top$. Then

$$v - W_{t+1} k = e - \alpha e \|k\|_2^2 = (1 - \alpha \|k\|_2^2) e.$$

The implicit step gives

$$\alpha e k^\top = \eta (1 - \alpha \|k\|_2^2) e k^\top.$$

Thus

$$\alpha = \frac{\eta}{1 + \eta \|k\|_2^2}.$$

Substituting α gives the update and residual contraction. □

Theorem D.7 (KL consistency projection)**.** *Let p_t be strictly positive on a finite space $\mathcal{X}$. The solution of*

$$\operatorname*{argmin}_{p:\mathbb{E}_p[\phi]=c} D_{\mathrm{KL}}(p\|p_t)$$

has the form

$$p_{t+1}(x) = \frac{p_t(x)\exp(\lambda^\top \phi(x))}{Z(\lambda)}.$$

Proof. The Lagrangian is

$$\mathcal{L}(p,\lambda,\nu) = \sum_x p(x)\log\frac{p(x)}{p_t(x)} + \lambda^\top\left(\sum_x p(x)\phi(x) - c\right) + \nu\left(\sum_x p(x) - 1\right).$$

Stationarity gives

$$\log\frac{p(x)}{p_t(x)} + 1 + \lambda^\top\phi(x) + \nu = 0.$$

Hence, after absorbing signs into λ and normalizing,

$$p_{t+1}(x) = \frac{p_t(x)\exp(\lambda^\top \phi(x))}{Z(\lambda)}.$$

□

Proposition D.8 (Gradient descent as linearized projection)**.** *For $D_\psi(s,s_t) = \|s-s_t\|_2^2/2$, replacing $\ell_t(s)$ in the soft projection objective by its first-order approximation at s_t gives $s_{t+1} = s_t - \eta\nabla\ell_t(s_t)$.*

Proof. The linearized objective is

$$\frac{1}{2}\|s-s_t\|_2^2 + \eta\langle\nabla\ell_t(s_t), s-s_t\rangle + \eta\ell_t(s_t).$$

The stationarity condition is

$$s - s_t + \eta\nabla\ell_t(s_t) = 0.$$

Thus

$$s_{t+1} = s_t - \eta\nabla\ell_t(s_t).$$

□

E Conservative Memory Proofs

Theorem E.1 (Exact associative binding, formal version of Theorem 3.4)**.** *Let $K \in \mathbb{R}^{r\times T}$ and $V \in \mathbb{R}^{m\times T}$. The set $\mathcal{C} = \{W \in \mathbb{R}^{m\times r} : WK = V\}$ is nonempty if and only if $\ker K \subseteq \ker V$. If $\mathcal{C}$ is nonempty, then*

$$\operatorname*{argmin}_{W\in\mathcal{C}} \frac{1}{2}\|W - W_0\|_F^2 = W_0 + (V - W_0K)K^\dagger.$$

Proof. If $WK = V$ and $a \in \ker K$, then

$$Va = WKa = 0.$$

Thus $\ker K \subseteq \ker V$.

Assume $\ker K \subseteq \ker V$. Since $K^\dagger K$ is the orthogonal projector onto $(\ker K)^\perp$, the condition implies

$$V = VK^\dagger K.$$

Thus $W = VK^\dagger$ satisfies $WK = V$.

Let $R = V - W_0 K$ and $W = W_0 + \Delta$. The projection problem is

$$\min_\Delta \frac{1}{2} \|\Delta\|_F^2 \quad \text{subject to} \quad \Delta K = R.$$

The minimum-Frobenius-norm solution is

$$\Delta^\star = RK^\dagger.$$

Therefore

$$W^\star = W_0 + (V - W_0 K)K^\dagger.$$

□

Theorem E.2 (Innovation decomposition, formal version of Theorem 3.5)**.** *Let $K \in \mathbb{R}^{r\times T}$, $V \in \mathbb{R}^{m\times T}$, $W \in \mathbb{R}^{m\times r}$, $k \in \mathbb{R}^r$, and $v \in \mathbb{R}^m$. Assume $WK = V$. Let $\Pi = KK^\dagger$, $z = (I - \Pi)k$, and $a = K^\dagger k$. Then*

$$Wk = Va + Wz.$$

If $z = 0$, Wk is fixed over all old-feasible W. If $z \neq 0$, the constraint $Wk = v$ is equivalent over old-feasible W to $Wz = v - Va$.

Proof. The decomposition is

$$k = KK^\dagger k + (I - KK^\dagger)k = Ka + z.$$

Multiplying by W gives

$$Wk = WKa + Wz = Va + Wz.$$

If $z = 0$, then $Wk = Va$ for every W satisfying $WK = V$. If $z \neq 0$, then

$$Wk = v \quad \Longleftrightarrow \quad Va + Wz = v \quad \Longleftrightarrow \quad Wz = v - Va.$$

□

Theorem E.3 (Conflict impossibility, formal version of Theorem 3.6)**.** *Let $K \in \mathbb{R}^{r\times T}$, $V \in \mathbb{R}^{m\times T}$, $W_t \in \mathbb{R}^{m\times r}$, $k \in \mathbb{R}^r$, and $v \in \mathbb{R}^m$. Assume $W_t K = V$ and $z = (I - KK^\dagger)k = 0$. If $W_t + \Delta$ preserves old evidence, then $(W_t + \Delta)k = W_t k$. Hence $Wk = v$ is compatible with old evidence if and only if $v = W_t k$.*

Proof. Preserving old evidence means

$$(W_t + \Delta)K = V = W_t K.$$

Thus

$$\Delta K = 0.$$

If $z = 0$, then $k \in \operatorname{span}(K)$, hence

$$\Delta k = 0.$$

Therefore

$$(W_t + \Delta)k = W_t k.$$

□

Theorem E.4 (Hard Conservative Projection Memory)**.** *Let $K \in \mathbb{R}^{r\times T}$, $V \in \mathbb{R}^{m\times T}$, $W_t \in \mathbb{R}^{m\times r}$, $k \in \mathbb{R}^r$, and $v \in \mathbb{R}^m$. Assume $W_t K = V$, let $\Pi = KK^\dagger$, $z = (I - \Pi)k \neq 0$, and $e = v - W_t k$. Then*

$$W_{t+1} = W_t + \frac{ez^\top}{\|z\|_2^2}$$

preserves $WK = V$, satisfies $Wk = v$, and minimizes $\|W - W_t\|_F^2 / 2$ among all memories satisfying both constraints.

Proof. Let $\Delta^\star = ez^\top / \|z\|_2^2$. Since $z \perp \operatorname{span}(K)$,

$$\Delta^\star K = 0.$$

Thus

$$W_{t+1}K = W_t K = V.$$

Also $z^\top k = \|z\|_2^2$, so

$$\Delta^\star k = e.$$

Hence

$$W_{t+1}k = W_t k + e = v.$$

For minimality, any feasible update Δ satisfies

$$\Delta K = 0, \qquad \Delta k = e.$$

Since $\Delta K = 0$,

$$\Delta k = \Delta(I - \Pi)k = \Delta z.$$

Thus the problem reduces to

$$\min_{\Delta} \frac{1}{2}\|\Delta\|_F^2 \quad \text{subject to} \quad \Delta z = e.$$

The minimum-Frobenius-norm solution is $\Delta^\star = ez^\top / \|z\|_2^2$. □

Theorem E.5 (No-free-repair lower bound, formal version of Theorem 3.8)**.** *Let $K \in \mathbb{R}^{r\times T}$, $V \in \mathbb{R}^{m\times T}$, $W_t \in \mathbb{R}^{m\times r}$, $k \in \mathbb{R}^r$, $v \in \mathbb{R}^m$, and $0 \le \gamma < 1$. Assume $W_t K = V$, let $\Pi = KK^\dagger$, $z = (I - \Pi)k$, and $e = v - W_t k$. Let $W_t + \Delta$ preserve old constraints, so $\Delta K = 0$. If $z \neq 0$ and*

$$\|v - (W_t + \Delta)k\|_2 \le \gamma \|e\|_2$$

then

$$\|\Delta\|_F \ge \frac{(1-\gamma)\|e\|_2}{\|z\|_2}.$$

If $z = 0$ and $e \neq 0$, no update preserving old constraints can achieve such a contraction.

Proof. Since $\Delta K = 0$,

$$\Delta k = \Delta(I - \Pi)k = \Delta z.$$

Let

$$e^+ = v - (W_t + \Delta)k.$$

Then

$$e^+ = e - \Delta z.$$

Thus

$$\|\Delta z\|_2 = \left\|e - e^+\right\|_2 \ge \|e\|_2 - \left\|e^+\right\|_2 \ge (1-\gamma)\|e\|_2 .$$

Also

$$\|\Delta z\|_2 \le \|\Delta\|_F \|z\|_2 .$$

If $z \neq 0$, division gives the result. If $z = 0$, then $\Delta z = 0$ and $e^+ = e$, so no contraction with $\gamma < 1$ is possible when $e \neq 0$. □

Theorem E.6 (Optimal contraction, formal version of Theorem 3.9)**.** *Let $K \in \mathbb{R}^{r\times T}$, $V \in \mathbb{R}^{m\times T}$, $W_t \in \mathbb{R}^{m\times r}$, $k \in \mathbb{R}^r$, $v \in \mathbb{R}^m$, and $0 \le \gamma \le 1$. Assume $W_t K = V$, let $\Pi = KK^\dagger$, $z = (I - \Pi)k \neq 0$, and $e = v - W_t k$. Among all updates satisfying*

$$\Delta K = 0, \qquad v - (W_t + \Delta)k = \gamma e,$$

the unique minimum-Frobenius-norm update is

$$\Delta^\star = (1-\gamma)\frac{ez^\top}{\|z\|_2^2}.$$

Its Frobenius norm is

$$\|\Delta^\star\|_F = \frac{(1-\gamma)\|e\|_2}{\|z\|_2}.$$

Proof. The residual constraint is

$$e - \Delta k = \gamma e.$$

Since $\Delta K = 0$, $\Delta k = \Delta z$. Thus

$$\Delta z = (1-\gamma)e.$$

The minimum-Frobenius-norm solution of this linear constraint is

$$\Delta^\star = (1-\gamma)\frac{ez^\top}{\|z\|_2^2}.$$

It preserves old constraints because $z^\top K = 0$:

$$\Delta^\star K = 0.$$

It achieves the target because

$$\Delta^\star z = (1-\gamma)e.$$

The Frobenius norm is

$$\|\Delta^\star\|_F = (1-\gamma)\frac{\|e\|_2 \|z\|_2}{\|z\|_2^2} = \frac{(1-\gamma)\|e\|_2}{\|z\|_2}.$$

□

Theorem E.7 (Sequential hard CPM equals batch projection, formal version of Theorem 3.10)**.** *Let $(k_t, v_t)_{t=1}^T$ be compatible evidence, and let $K_t = [k_1, \dots, k_t]$ and $V_t = [v_1, \dots, v_t]$. Initialize $W_1 = W_0$. At step t, let $\Pi_{t-1} = K_{t-1}K_{t-1}^\dagger$ and $z_t = (I - \Pi_{t-1})k_t$. If $z_t = 0$, make no update. If $z_t \neq 0$, apply hard CPM:*

$$W_{t+1} = W_t + \frac{(v_t - W_t k_t)z_t^\top}{\|z_t\|_2^2}.$$

Then

$$W_{T+1} = \operatorname*{argmin}_{W: WK_T = V_T} \frac{1}{2}\|W - W_0\|_F^2.$$

Proof. Let

$$\mathcal{C}_t = \{W : WK_t = V_t\}, \qquad \mathcal{T}_t = \{H : HK_t = 0\}.$$

We prove by induction that W_{t+1} is the Frobenius projection of W_0 onto $\mathcal{C}_t$. For $t = 0$, this is $W_1 = W_0$.

Assume $W_t = \mathrm{Proj}_{\mathcal{C}_{t-1}}(W_0)$. Thus $W_t \in \mathcal{C}_{t-1}$ and

$$\langle H, W_t - W_0\rangle_F = 0 \qquad \forall H \in \mathcal{T}_{t-1}.$$

If $z_t = 0$, compatibility implies the new constraint is already implied by $\mathcal{C}_{t-1}$, hence $\mathcal{C}_t = \mathcal{C}_{t-1}$ and no update preserves the projection.

If $z_t \neq 0$, hard CPM gives $W_{t+1} \in \mathcal{C}_t$ by Theorem E.4. Let $H \in \mathcal{T}_t$. Then $H \in \mathcal{T}_{t-1}$ and $Hk_t = 0$. Since $\Pi_{t-1}k_t \in \operatorname{span}(K_{t-1})$,

$$Hz_t = Hk_t - H\Pi_{t-1}k_t = 0.$$

Let $\Delta_t = W_{t+1} - W_t$. Then

$$\langle H, \Delta_t \rangle_F = \left\langle H, (v_t - W_t k_t) z_t^\top \right\rangle_F \frac{1}{\|z_t\|_2^2} = \frac{(v_t - W_t k_t)^\top H z_t}{\|z_t\|_2^2} = 0.$$

Therefore

$$\langle H, W_{t+1} - W_0 \rangle_F = \langle H, W_t - W_0 \rangle_F + \langle H, \Delta_t \rangle_F = 0.$$

The Frobenius projection characterization onto the affine set $\mathcal{C}_t$ gives $W_{t+1} = \operatorname{Proj}_{\mathcal{C}_t}(W_0)$. □

Theorem E.8 (Soft Conservative Projection Memory, formal version of Theorem 3.7)**.** *Let $K \in \mathbb{R}^{r\times T}$, $V \in \mathbb{R}^{m\times T}$, $W_t \in \mathbb{R}^{m\times r}$, $k \in \mathbb{R}^r$, $v \in \mathbb{R}^m$, and $\eta > 0$. Assume $W_t K = V$, let $\Pi = KK^\dagger$, $z = (I - \Pi)k \neq 0$, and $e = v - W_t k$. The unique solution of*

$$\operatorname*{argmin}_{W:WK=V} \left\{ \frac{1}{2} \|W - W_t\|_F^2 + \frac{\eta}{2} \|Wk - v\|_2^2 \right\}$$

is

$$W_{t+1} = W_t + \frac{\eta}{1 + \eta \|z\|_2^2} e z^\top.$$

Moreover,

$$W_{t+1}K = V, \qquad v - W_{t+1}k = \frac{1}{1 + \eta \|z\|_2^2} e.$$

Proof. Write $W = W_t + \Delta$. The constraint $WK = V$ is equivalent to

$$\Delta K = 0.$$

Since $\Delta K = 0$,

$$\Delta k = \Delta(I - \Pi)k = \Delta z.$$

The objective becomes

$$\min_{\Delta:\Delta K=0} \frac{1}{2} \|\Delta\|_F^2 + \frac{\eta}{2} \|\Delta z - e\|_2^2 .$$

The stationarity equation in the feasible subspace is

$$\Delta + \eta(\Delta z - e)z^\top = 0.$$

Let $\Delta = \alpha e z^\top$. Substitution gives

$$\alpha e z^\top + \eta(\alpha e \|z\|_2^2 - e)z^\top = 0.$$

Thus

$$\alpha(1 + \eta \|z\|_2^2) = \eta.$$

Therefore

$$\Delta = \frac{\eta}{1 + \eta \|z\|_2^2} e z^\top.$$

Since $z^\top K = 0$, $W_{t+1} K = V$. The residual is

$$\begin{aligned} v - W_{t+1} k &= e - \Delta k \\ &= e - \Delta z \\ &= e - \frac{\eta \|z\|_2^2}{1 + \eta \|z\|_2^2} e \\ &= \frac{1}{1 + \eta \|z\|_2^2} e. \end{aligned}$$

□

F Ridge and Bayesian Projection Memory Proofs

Theorem F.1 (Ridge solution)**.** *Let $W_0 \in \mathbb{R}^{m \times r}$, $\lambda > 0$, $\rho_\tau \geq 0$, $k_\tau \in \mathbb{R}^r$, and $v_\tau \in \mathbb{R}^m$. Let*

$$J_t(W) = \frac{\lambda}{2} \|W - W_0\|_F^2 + \frac{1}{2} \sum_{\tau=1}^{t} \rho_\tau \|W k_\tau - v_\tau\|_2^2.$$

Define

$$A_t = \lambda I + \sum_{\tau=1}^{t} \rho_\tau k_\tau k_\tau^\top, \qquad B_t = \lambda W_0 + \sum_{\tau=1}^{t} \rho_\tau v_\tau k_\tau^\top.$$

Then the unique minimizer is $W_{t+1} = B_t A_t^{-1}$.

Proof. Differentiate:

$$\nabla J_t(W) = \lambda(W - W_0) + \sum_{\tau=1}^{t} \rho_\tau (W k_\tau - v_\tau) k_\tau^\top.$$

At optimum,

$$0 = \lambda(W - W_0) + \sum_{\tau=1}^{t} \rho_\tau (W k_\tau - v_\tau) k_\tau^\top.$$

Thus

$$W \left(\lambda I + \sum_{\tau=1}^{t} \rho_\tau k_\tau k_\tau^\top \right) = \lambda W_0 + \sum_{\tau=1}^{t} \rho_\tau v_\tau k_\tau^\top.$$

Since $\lambda > 0$, A_t is invertible. Hence

$$W_{t+1} = B_t A_t^{-1}.$$

□

Theorem F.2 (Recursive ridge update)**.** *Under the notation of Theorem F.1, let* $P_t = A_{t-1}^{-1}$ *and* $W_t = B_{t-1}P_t$*. Define*

$$g_t = \frac{\rho_t P_t k_t}{1 + \rho_t k_t^\top P_t k_t}.$$

Then

$$W_{t+1} = W_t + (v_t - W_t k_t) g_t^\top, \qquad P_{t+1} = P_t - \frac{\rho_t P_t k_t k_t^\top P_t}{1 + \rho_t k_t^\top P_t k_t}.$$

Proof. The Sherman-Morrison identity gives

$$P_{t+1} = A_t^{-1} = P_t - \frac{\rho_t P_t k_t k_t^\top P_t}{1 + \rho_t k_t^\top P_t k_t}.$$

Let $a_t = k_t^\top P_t k_t$. Then

$$k_t^\top P_{t+1} = k_t^\top P_t - \frac{\rho_t a_t k_t^\top P_t}{1 + \rho_t a_t} = \frac{k_t^\top P_t}{1 + \rho_t a_t}.$$

Using $B_t = B_{t-1} + \rho_t v_t k_t^\top$,

$$\begin{aligned} W_{t+1} &= B_t P_{t+1} \\ &= B_{t-1} P_{t+1} + \rho_t v_t k_t^\top P_{t+1} \\ &= W_t - \frac{\rho_t W_t k_t k_t^\top P_t}{1 + \rho_t a_t} + \frac{\rho_t v_t k_t^\top P_t}{1 + \rho_t a_t} \\ &= W_t + (v_t - W_t k_t) \frac{\rho_t k_t^\top P_t}{1 + \rho_t a_t} \\ &= W_t + (v_t - W_t k_t) g_t^\top. \end{aligned}$$

□

Theorem F.3 (Ridge bound, formal version of Theorem 3.12)**.** *Let* $W_0 \in \mathbb{R}^{m \times r}$*,* $\lambda > 0$*,* $\rho_t \geq 0$*,* $k_t \in \mathbb{R}^r$*, and* $v_t \in \mathbb{R}^m$*. With* J_t *and* A_t *as in Theorem F.1, let*

$$W_t = \operatorname*{argmin}_{W \in \mathbb{R}^{m \times r}} J_{t-1}(W), \qquad P_t = A_{t-1}^{-1}, \qquad a_t = k_t^\top P_t k_t, \qquad e_t = v_t - W_t k_t.$$

For every $U \in \mathbb{R}^{m \times r}$*,*

$$\sum_{t=1}^{T} \frac{\rho_t}{1 + \rho_t a_t} \|e_t\|_2^2 \leq \lambda \|U - W_0\|_F^2 + \sum_{t=1}^{T} \rho_t \|v_t - U k_t\|_2^2.$$

Proof. Since W_t minimizes J_{t-1}, for any Δ,

$$J_{t-1}(W_t + \Delta) = J_{t-1}(W_t) + \frac{1}{2} \operatorname{tr}(\Delta A_{t-1} \Delta^\top).$$

Also

$$J_t(W_t + \Delta) = J_{t-1}(W_t + \Delta) + \frac{\rho_t}{2} \|(W_t + \Delta) k_t - v_t\|_2^2.$$

With $e_t = v_t - W_t k_t$, the increment minimization is

$$\min_{\Delta} \left\{ \frac{1}{2} \operatorname{tr}(\Delta A_{t-1} \Delta^\top) + \frac{\rho_t}{2} \|\Delta k_t - e_t\|_2^2 \right\}.$$

Solving row-wise gives minimum value

$$\frac{\rho_t}{2(1+\rho_t a_t)} \|e_t\|_2^2 .$$

Hence

$$J_t(W_{t+1}) - J_{t-1}(W_t) = \frac{\rho_t}{2(1+\rho_t a_t)} \|e_t\|_2^2 .$$

Summing over t and using $W_1 = W_0$ gives

$$\sum_{t=1}^{T} \frac{\rho_t}{2(1+\rho_t a_t)} \|e_t\|_2^2 = J_T(W_{T+1}).$$

Since W_{T+1} minimizes J_T,

$$J_T(W_{T+1}) \le J_T(U) = \frac{\lambda}{2} \|U - W_0\|_F^2 + \frac{1}{2} \sum_{t=1}^{T} \rho_t \|U k_t - v_t\|_2^2 .$$

Multiplying by 2 proves the claim. □

Theorem F.4 (Log-determinant information identity, formal version of Theorem 3.13)**.** *Let $\lambda > 0$, $\rho_t \ge 0$, $k_t \in \mathbb{R}^r$, $A_t = \lambda I + \sum_{\tau=1}^{t} \rho_\tau k_\tau k_\tau^\top$, and $a_t = k_t^\top A_{t-1}^{-1} k_t$. Then*

$$\sum_{t=1}^{T} \log(1+\rho_t a_t) = \log \frac{\det A_T}{\det A_0}.$$

Proof. The matrix determinant lemma gives

$$\begin{aligned} \det A_t &= \det(A_{t-1} + \rho_t k_t k_t^\top) \\ &= \det A_{t-1}(1 + \rho_t k_t^\top A_{t-1}^{-1} k_t) \\ &= \det A_{t-1}(1 + \rho_t a_t). \end{aligned}$$

Taking logarithms,

$$\log\det A_t - \log\det A_{t-1} = \log(1+\rho_t a_t).$$

Summing over t gives the identity. □

Theorem F.5 (Bayesian ridge projection memory, formal version of Theorem 3.11)**.** *Let $W \in \mathbb{R}^{m\times r}$ and $\rho_t > 0$. Assume each row w_j of W has prior $w_j \sim \mathcal{N}(w_{0,j}, \lambda^{-1} I)$ and observations satisfy $v_t = W k_t + \varepsilon_t$ with $k_t \in \mathbb{R}^r$, $v_t \in \mathbb{R}^m$, and $\varepsilon_t \sim \mathcal{N}(0, \rho_t^{-1} I_m)$ independently over time. Then the posterior mean and MAP estimate are $B_t A_t^{-1}$.*

Proof. The negative log posterior, up to constants independent of W, is

$$\frac{\lambda}{2}\|W - W_0\|_F^2 + \frac{1}{2}\sum_{\tau=1}^{t}\rho_\tau \|Wk_\tau - v_\tau\|_2^2 .$$

This is $J_t(W)$. The posterior is Gaussian because the prior is Gaussian and the observation model is linear Gaussian. Hence the posterior mean equals the MAP. By Theorem F.1, this value is

$$B_t A_t^{-1}.$$

□